

Chunxiao Jing

+1-608-504-8549 | cjing7@wisc.edu

Agricultural and Applied Economics, University of Wisconsin-Madison
216 Taylor Hall, 427 Lorch Street, Madison, WI 53706

EDUCATION

- **University of Wisconsin-Madison** *August, 2021-May, 2027(expected)*
Ph.D., Agricultural and Applied Economics Madison, US
- **University of California, Los Angeles** *June, 2019*
M.A., Applied Economics Los Angeles, US
- **Nankai University** *June, 2018*
B.A., Economics Tianjin, China

RESEARCH INTEREST

Development Economics, Structural Transformation in Africa and China, Labor Economics, Climate Change, Machine Learning

SKILLS

- **Computer Skills:** Python | Stata | R | $\LaTeX$ | QGIS | SPSS
- **Certificates:** Bloomberg Certification(US) | Certificate of Level 2 in National Computer Rank Examination(China) | Certificate of Futures Qualification(China) | Certificate of Securities(China) | Certificate of Accounting Profession(China)

WORKING PAPER

- **Can the Service Sector Lead Structural Transformation in Africa? Evidence from Côte d'Ivoire**
with Jeremy Foltz
 - Estimated firm-level total factor productivity in agriculture, manufacturing, and service using the MrEst Method
 - Analyzed the movement of skilled and unskilled workers across firms with varying productivity levels
 - Suggested that policymakers across Africa reconsider a manufacturing-first development strategy and balance their efforts between manufacturing and services
- **The Impact of FDI Policy on Different Ownership Structure Enterprises in China**
 - Examined changes in the market power of Chinese domestic private-owned enterprises, joint ventures, and foreign-owned enterprises before and after the 2002 FDI policy
 - Calculated firm-level markups and analyzed the FDI policy's impact on firms' market power
 - Suggested that policymakers consider differences in firm ownership, as the effects of the FDI policy vary by ownership type

WORK IN PROGRESS

- **Structural Transformation Under Subsistence Constraints and Entry Cost: Evidence from Weather Productivity in Africa**
 - This paper studies how agricultural productivity shocks translate into structural transformation in Sub-Saharan Africa. I combine nationally representative household panel data from Ethiopia, Malawi, and Niger with high-resolution climate data to construct a machine-learning-based measure of weather-driven agricultural productivity. The results show that productivity shocks significantly affect labor reallocation, but their effects vary across countries due to movement barriers such as subsistence constraints, spatial frictions, and land policy restrictions. To explain these differences, I develop and estimate a structural model of household labor allocation. Counterfactual simulations evaluate the roles of food subsidies, productivity improvements, land policy reform, and rural road investments in promoting labor reallocation and economic development under climate-driven shocks.
- **Climate Change and Conflict: Causal Evidence using New Machine Learning Methods**
with Jeremy Foltz
 - This paper examines whether climate variability can predict armed conflict in Africa. Using grid-cell data from 2004 to 2020 that combine conflict events with climate indicators and land-cover information, we compare logistic regression with machine-learning methods, including Random Forest and XGBoost. The machine-learning models achieve higher predictive recall rate (96%), with a more balanced recall and precision values under a five-year rolling window. To move beyond prediction, we open the black box of machine learning by applying the top predictive features to causal inference. Specifically, we classify treatment into three exposure categories and estimate direct-high, direct-low, and spillover using double debiased machine learning (DDML) for treatment effects. The analysis reveals distinct spatial mechanisms: spillover effects across cells are prevalent, conflict risk intensifies under widespread exposure as regional scarcity or disruption limits adaptation (Direct-High), but declines when shocks are localized and coping through migration or trade is possible (Direct-Low).

PROFESSIONAL EXPERIENCE

- **Research Assistant for Professor Jeremy Foltz**

University of Wisconsin-Madison, Department of Agricultural and Applied Economics

2024-2025

- Collected commune-level NDVI data for Mali from Google Earth Engine using Python
- Analyzed the correlation between climate change and conflicts in Mali using R and Stata

- **Research Assistant for Professor Jeremy Foltz**

University of Wisconsin-Madison, Department of Agricultural and Applied Economics

2022-2024

- Estimated firm-level total factor productivity in agriculture, manufacturing, and services using various proxy variable techniques (LP, OP, ACF, and MrEst Methods) in Côte d'Ivoire
- Wrote a paper on labor reallocation patterns across sectors

- **Research Assistant in MAE Research Lab**

University of California Los Angeles, Department of Economics

Spring, 2019

- Conducted literature reviews on Google Trends prediction
- Downloaded query searches from Google Trends using Google Correlate
- Applied machine learning methods in Python and R to predict unemployment in the US

TEACHING EXPERIENCE

- **The Growth and Development of Nations in the Global Economy (AAE 374)**

Taught by Professor Vladimir Gimpelson

Teaching Assistant

Spring, 2024

- **The Growth and Development of Nations in the Global Economy (AAE 374)**

Taught by Professor Jeremy Foltz

Teaching Assistant

Spring, 2023

CONFERENCES AND PRESENTATIONS

* INDICATES UPCOMING

2026 MEA Annual Meeting, AAEA Annual Meeting*

2025 ASSA Annual Meeting, AAE Development Workshop

2024 CSAE Conference, CES China Annual Conference, AAEA Annual Meeting

2023 AAE Development Workshop

FUNDINGS AND GRANTS

STEG Small Research Grant, 2026-2027, £5,000

STEG PhD Research Grant, 2026-2027, £10,000

UW-Madison PhD Fellowship, 2025-2026, \$35,636

AAE Conference Travel Grant, 2024-2025, \$1,000

AAE Conference Travel Grant, 2023-2024, \$1,000

UW-Madison PhD Fellowship, 2021-2022, \$30,545

PROFESSIONAL SERVICE

- **Mentor Service**

Mentor, The Graduate Applications International Network (GAIN) Program

2022-2023 and 2024-2025

- **Department Service**

The Taylor-Hibbard Club President, University of Wisconsin-Madison

2023-2024

ADDITIONAL INFORMATION

Languages: Chinese (Native), English (Fluent)

Interests: Swimming, Sailing, Gardening

REFERENCES

Jeremy Foltz

Professor

Department of Agricultural and Applied Economics

University of Wisconsin-Madison

Email: jdfoltz@wisc.edu

Vladimir Gimpelson

Professor

Centre for Labour Market Studies

Higher School of Economics

Email: vladim.gimpelson@gmail.com